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Q1.

The graph ~~has~~ shown has:

→ Domain from $-\frac{1}{2}$ to $\frac{1}{2}$

→ Range from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$

→ The graph has the shape of an inverse sine fn.

So, for $y = \sin^{-1} x$, the domain is $-1 \leq x \leq 1$

But here, graph is horizontally compressed.

So, Domain Becomes.

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

This happens for,

$$y = \sin^{-1}(2x)$$

because for $\sin^{-1}(2x)$,

$$-1 \leq 2x \leq 1$$

So,

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

$\therefore$

$$y = \sin^{-1}(2x)$$

ans.

Q2.

Given,

$$A = [a_{ij}]_{m \times 4}, B = [b_{ij}]_{n \times 3}$$

$$C = [c_{ij}]_{p \times q}$$

So, For $A * B$ to be defined,

$$\text{cols of } A = \text{rows of } B$$

$$\therefore 4 = n$$
$$\Rightarrow m = 4$$

Order of $AB \Rightarrow m \times 3$

But $A B$ is a square matrix,

$$\text{So, } m = 3$$

$$\text{Thus, } m = 3, n = 4$$

$$\text{Similarly, } p = 4$$

$$\Rightarrow q = 3$$

$$\boxed{m = q = 3, n = p = 4}$$